

Nonsmooth Finite-Time Synchronization of Switched Coupled Neural Networks

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Abstract—This paper is concerned with the finite-time synchronization (FTS) issue of switched coupled neural networks with discontinuous or continuous activations. Based on the framework of nonsmooth analysis, some discontinuous or continuous controllers are designed to force the coupled networks to synchronize to an isolated neural network. Some sufficient conditions are derived to ensure the FTS by utilizing the well-known finite-time stability theorem for nonlinear systems. Compared with the previous literatures, such synchronization objective will be realized when the activations and the controllers are both discontinuous. The obtained results in this paper include and extend the earlier works on the synchronization issue of coupled networks with Lipschitz continuous conditions. Moreover, an upper bound of the settling time for synchronization is estimated. Finally, numerical simulations are given to demonstrate the effectiveness of the theoretical results.

Index Terms—Discontinuous activations, finite-time synchronization (FTS), neural networks, switched coupling.

I. INTRODUCTION

IT IS well known that extensive fundamental results have been reported to address the dynamic behaviors of neural networks, such as stability, convergence, and synchronization [1]–[5]. However, a great deal of results is based on the assumption that the activation functions are continuous, Lipschitzian, or even continuously differentiable [6]. Generally, due to the conventional definition and simple existence conditions of solutions, some smoothness assumptions have been introduced in analyzing or controlling the considered systems. Nevertheless, in mathematics and control theory or in various science and engineering applications,

some system dynamics are discontinuous, such as dry friction, impacting machines, and power circuits [7]. Similarly, neural networks with discontinuous activation functions also frequently arise in practice [8], which can reveal important traits of the dynamics such as the phenomenon of converging toward the limit cycle in a finite time and the capability of calculating the global minimum of the underlying energy function [9].

In the past few decades, the synchronization problem of complex networks has become a hot topic due to its potential applications. Synchronization, which is indeed a basis to understand an unknown dynamic system from one or more well-known dynamic systems. To put it in another way, the response complexity of an unknown system to one or more well-known systems can be measured and compared through such synchronicity [10]–[12]. For a given network, its behavior is determined by two main features: 1) the dynamics of each isolated nodes and 2) the connections between them. However, it is not always practical to assume that the connections between each pair of nodes are always time-invariant since in some real-world complex networks the link failure and new link creation may happen as time evolves [13], [14]. Hence, switches of network structure between some different topologies are inevitable for many dynamic networks. Naturally, the synchronization of switched networks has been paid considerable research attention in this paper due to their numerous applications in many fields [15].

Recently, some work studied the synchronization of switched coupled networks (see [16]–[20]). To mention a few, Liu *et al.* [18] gave some sufficient condition for almost sure synchronization of the networks with two kinds of stochastic switched coupling. Tang *et al.* [19] investigated the distributed synchronization problem of networks of agent with controllers and nonlinearities subject to Bernoulli switchings and found that some factors had impacts on the convergence speed and the terminal control strength; Wu *et al.* [20] studied the problems of exponential stability analysis and synchronization of discrete-time switched delayed neural networks by using the average dwell time approach together with the piecewise Lyapunov function technique. Notice that all the above-mentioned results only considered exponential or asymptotical convergence. However, finite-time convergence is more valuable in practical application due to the advantage of faster convergence rate and robustness against uncertainties. Therefore, one common question could be: how to ensure finite-time synchronization (FTS) for the switched coupled neural

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networks (SCNNs). This question motivates us to carry out this paper.

Up to date, there has been an increasing interest in considering FTS (including consensus) [21]–[35], where the ideas and methods are mainly based on finite-time stability theory for both continuous systems [33] and discontinuous systems [34]. Taking some literatures, for example, Yang and Cao [21] studied the finite-time stochastic synchronization by designing a continuous controller. Shen and Cao [22] investigated FTS of an array of coupled neural networks with discontinuous controllers. The finite-time distributed consensus problem for multiagent systems was considered in [30] by using a (discontinuous) binary consensus protocol and a pinning control scheme, where only the sign information of the relative state measurements was required.

Unfortunately, the FTS of SCNNs with discontinuous activations has not been fully investigated: how to deal with the switched system, discontinuous or continuous activations, and finite-time control in a unified framework; how to ensure the FTS of switched neural networks when the activations and controllers are both discontinuous; and how to explain the result of this synchronization problem could include the previous researches are remaining challenges.

Compared with the existing results about FTS, this paper considers some more general cases, and the novelties lie in several aspects. First, the problem of FTS for SCNNs will be solved when the activations and controllers are both discontinuous. That is, by designing some discontinuous controllers, FTS can always be ensured under some conditions whether the activations are continuous or not. Second, for the coupled neural networks with continuous activations, a discontinuous pinning controller is further designed to guarantee FTS, where a single controller can successfully pin a coupled network to a given isolated neural network as well. Third, it is shown that continuous controllers can still ensure synchronization when the activations are continuous, which includes and extends the earlier results about FTS for coupled neural networks. Finally, all the above-mentioned results are based on the switched network structure, and the upper-bounds of the settling time are all expressed explicitly as well.

II. PRELIMINARIES AND MODEL FORMULATION

A. Filippov Solutions and Set-Valued Lie Derivative

Let $E \subset \mathbb{R}^n$ be a subset. Map $x \mapsto \mathbb{F}(x)$ is called a set-valued map from $E \rightarrow \mathcal{B}(\mathbb{R}^n)$, where $\mathcal{B}(\mathbb{R}^n)$ is the set consisting of all the subsets of $\mathbb{R}^n$, if to each point x of the set E , there corresponds a nonempty set $\mathbb{F}(x) \subset \mathbb{R}^n$.

For a system $(dx/dt) = f(x)$, even if $f(\cdot)$ is not continuous, the Filippov set-valued map of f (also called Filippov regularization [36]) is defined as $\mathbb{F}(x) = \bigcap_{\delta>0} \bigcap_{\mu(N)=0} K[f(B(x, \delta) \setminus N)]$, where $K(E)$ is the closure of the convex hull of set E , $B(x, \delta) = \{y : \|y - x\| \leq \delta\}$, and $\mu(M)$ is the Lebesgue measure of set $M \subset \mathbb{R}^n$. A solution of Filippov of this discontinuous system with the initial conditions $x(0) = x_0$ is an absolutely continuous function $x(t)$, $t \in [0, T]$, which satisfies $x(0) = x_0$ and differential inclusion: $\dot{x} \in \mathbb{F}(x(t))$; a.e. $t \in [0, T]$.

Let $\Omega_V \subset \mathbb{R}^n$ denotes the set of points at which function V fails to be differentiable. The generalized gradient [37] $\partial V : \mathbb{R}^n \rightarrow \mathcal{B}(\mathbb{R}^n)$ of V is defined by $\partial V(x) = \text{co}[\lim_{i \rightarrow \infty} \nabla V(x_i) : x_i \rightarrow x, x_i \notin S \cup \Omega_V]$, where $\text{co}(\cdot)$ denotes the convex hull and $S \subset \mathbb{R}^n$ is a set of measure zero.

The set-valued Lie derivative of V with respect to f at x is defined as $L_f V(x) = \{a \in \mathbb{R} \mid \exists v \in \mathbb{F}(x) \text{ with } v^T \xi = a, \forall \xi \in \partial V(x)\}$, where v is a measurable selection of $\mathbb{F}(x)$ [36].

B. Problem Formulation

The neural network model considered in this paper is described as follows:

$$\dot{x}(t) = -Dx(t) + Bf(x(t)) + J(t) \quad (1)$$

where $x(t) = (x_1(t), x_2(t), \dots, x_n(t))^T \in \mathbb{R}^n$ is the vector of neuron states; n represents the number of neurons in the network; $D = \text{diag}(d_1, d_2, \dots, d_n)$ is an $n \times n$ constant diagonal matrix with $d_i > 0, i = 1, 2, \dots, n$; $B = (b_{ij})_{n \times n}$ is an $n \times n$ interconnection matrix; $f(x) = (f_1(x_1), f_2(x_2), \dots, f_n(x_n))^T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a diagonal mapping where $f_i, i = 1, 2, \dots, n$, represents the neuron input–output activation and $J = (J_1, J_2, \dots, J_n)^T$ is a constant external input vector.

Unlike the previous studies on dynamics of neural networks, the nonlinear activation function f in this paper may not hold the Lipschitz condition [6], [13] or QUAD condition [39] any more. One of the remarkable differences between this paper and the existing work is that the node dynamics in our model are admitted to be discontinuous.

Definition 1: Class $\mathcal{F}$ of functions, we call $f \in \mathcal{F}$, if for all $i = 1, 2, \dots, n$, $f_i(\cdot)$ satisfies: $f_i(\cdot)$ is continuously differentiable, except on a countable set of isolated points $\{\rho_k^i\}$, where the right and left limits $f_i^+(\rho_k^i)$ and $f_i^-(\rho_k^i)$ exist, $k = 1, 2, \dots$.

Applying the framework of Filippov [36] in discussing the solution of neural network (1) with the discontinuous function $f \in \mathcal{F}$, we have the following differential inclusion:

$$\dot{x}(t) \in -Dx(t) + B\mathbb{F}(x) + J(t) \quad (2)$$

where $\mathbb{F}(x) \triangleq K[f(x)] = (K[f_1(x)], \dots, K[f_n(x)])$, and $K[f_i(x)] = [\min\{f_i(x^-), f_i(x^+)\}, \max\{f_i(x^-), f_i(x^+)\}]$, $i = 1, 2, \dots, n$.

Obviously, for all $f \in \mathcal{F}$, the above set-valued map $x(t) \mapsto -Dx(t) + B\mathbb{F}(x) + J(t)$ has nonempty compact convex values. Hence, it is upper-semi-continuous [40] and is measurable. By the measurable selection theorem [41], if $x(t)$ is a solution of (1), then there exists a measurable function $\beta(t) \in K[f(x(t))]$ such that for a.e. $t \in [0, +\infty)$, the following equation holds:

$$\dot{x}(t) = -Dx(t) + B\beta(t) + J(t), \text{ for a.e. } t \in [0, +\infty). \quad (3)$$

The neural network (1) with the switching signal can be described as follows:

$$\dot{x}(t) = -D_\pi x(t) + B_\pi f(x(t)) + J(t) \quad (4)$$

where π is a switching signal.

In order to study synchronization, we consider the following N SCNNs:

$$\dot{z}_i(t) = -D_\pi z_i(t) + B_\pi f(z_i(t)) + J(t) + c \sum_{j=1}^N a_{\pi ij} (z_j - z_i) \quad (5)$$

where $z_i(t) = (z_{i1}(t), z_{i2}(t), \dots, z_{in}(t))^T \in \mathbb{R}^n$ ($i = 1, 2, \dots, N$) is the state vector representing the state variables of node i at time t , c is the coupling strength, and π is a switching signal (including both the individual node and network topology switchings) which takes its value in the finite set $\pi = \{1, 2, \dots, M\}$. That is, at particular time, the matrices (D_π, B_π, A_π) are allowed to take values in a finite set $\{(D_1, B_1, A_1), (D_2, B_2, A_2), \dots, (D_M, B_M, A_M)\}$. Throughout this paper, the instantaneous value of switching rule π is assumed to be available in real time and unknown *a priori*. For every $\pi \in \{1, 2, \dots, M\}$, $A_\pi = [a_{\pi ij}]$ represents the adjacency matrix of the subsystems, and all of them are undirected, and its corresponding Laplacian matrix is denoted as L_π .

Remark 1: From the construction of the SCNNs (5), one can see that the switching information is unknown, but its instantaneous value is available in the real time. In fact, the complete synchronization may be ensured for SCNNs when the switched information is partially available. But in this paper, we aim to study the FTS issue for SCNNs, and the major concern is how to ensure FTS when the activations and controllers are all discontinuous. The problem concerning robust FTS with partially available switched information will be considered in our future research.

It is known that sometimes suitable nonlinear coupling can improve the network performance, such as in [42], where the synchronization for complex networks is realized via a kind of nonlinear coupling. In this paper, we will consider the FTS issue for the switched nonlinear coupled neural networks with discontinuous and continuous controllers, which can be formulated as follows:

$$\begin{aligned} \dot{z}_i(t) = & -D_\pi z_i(t) + B_\pi f(z_i(t)) + J(t) \\ & + c \sum_{j=1}^N a_{\pi ij} [z_j - z_i + \chi \text{sign}(z_j - z_i)] + u_{\pi i}(t) \end{aligned} \quad (6)$$

where χ is the nonlinear couple strength. That is, by adding a suitable designed controller $u_{\pi i}(t)$ ($i = 1, 2, \dots, N$) to networks, there exists a time T , one has $z_1(t) = z_2(t) = \dots = z_N(t) = s(t)$ when $t > T$.

Now, we define an indicator function $\bar{h}(t) = (\bar{h}_1(t), \bar{h}_2(t), \dots, \bar{h}_M(t))^T$, where

$$\bar{h}_q(t) = \begin{cases} 1 & \text{when the switched system is described} \\ & \text{by the } q\text{th mode } (D_q, B_q, A_q, u_q) \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

with $q = 1, 2, \dots, M$. It follows that $\sum_{q=1}^M \bar{h}_q(t) = 1$ under any switching rules.

Then, for $i = 1, 2, \dots, N$, the model of controlled switched coupled networks (6) can be rewritten as

$$\begin{aligned} \dot{z}_i(t) = & \sum_{q=1}^M \bar{h}_q(t) \left\{ -D_q z_i(t) + B_q f(z_i(t)) + J(t) \right. \\ & + c \sum_{j=1}^N a_{qij} [z_j - z_i + \chi \text{sign}(z_j - z_i)] \\ & \left. + u_{qi}(t) \right\}. \end{aligned} \quad (8)$$

III. MAIN RESULTS

In this section, we establish some basic results on the issue of FTS for SCNNs (6) by using some techniques from nonsmooth analysis. The main results are divided into two parts. In Part A, such issue is studied with discontinuous controllers. The same issue is further discussed with continuous controllers in Part B. And both of discontinuous and continuous activations two cases will be considered in each part, respectively.

The following lemmas are needed for the derivation of our main results in this paper.

Lemma 1 [33], [44]: Suppose that $V(x(t)) : \mathbb{R}^n \rightarrow \mathbb{R}$ is C -regular, and that $x(t) : [0, +\infty) \rightarrow \mathbb{R}^n$ is absolutely continuous on any compact interval of $[0, +\infty)$. If there exists a continuous function $\gamma : (0, +\infty) \rightarrow \mathbb{R}$, with $\gamma(\sigma) > 0$ for $\sigma \in (0, +\infty)$, such that

$$\dot{V}(t) \leq -\gamma(V(t))$$

and

$$\int_0^{V(0)} \frac{1}{\gamma(\sigma)} d\sigma = t^* < +\infty.$$

Then, we have $V(t) = 0$ for $t \geq t^*$. In particular:

- 1) if $\gamma(\sigma) = K_1\sigma + K_2\sigma^\mu$, for all $\sigma \in (0, +\infty)$, where $\mu \in (0, 1)$ and $K_1, K_2 > 0$, then the settling time is estimated by

$$t^* = \frac{1}{K_1(1-\mu)} \ln \frac{K_1 V^{1-\mu}(0) + K_2}{K_2} \quad (9)$$

- 2) if $\gamma(\sigma) = K\sigma^\mu$ and $K > 0$, then the settling time is estimated by

$$t^* = \frac{V^{1-\mu}(0)}{K(1-\mu)}. \quad (10)$$

Lemma 2 [45]: If $a_1, a_2, \dots, a_n$ are positive number and $0 < r < p$, then

$$\left(\sum_{i=1}^n a_i^p \right)^{1/p} \leq \left(\sum_{i=1}^n a_i^r \right)^{1/r}.$$

Lemma 3 [46]: Given an undirected graph $\mathcal{G}$ with adjacency matrix $\mathcal{A} = [a_{ij}]$ and Laplacian matrix L , equality

$$x^T L x = \frac{1}{2} \sum_{i,j=1}^n a_{ij} (x_i - x_j)^2$$

holds for arbitrary $x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$.

Lemma 4 [47]: If $A = [a_{ij}] \in \mathbb{R}^{m \times m}$ is an irreducible matrix with $\text{Rank}(A) = m - 1$ and satisfying $a_{ij} = a_{ji} \geq 0$, if $i \neq j$, and $\sum_{j=1}^m a_{ij} = 0$, for $i = 1, 2, \dots, m$. Let $\tilde{A} = [\tilde{a}_{ij}]$ with $\tilde{a}_{11} = a_{11} - \varepsilon$ ($\varepsilon > 0$) and $\tilde{a}_{ij} = a_{ij}$ otherwise. Then all eigenvalues of the matrix $\tilde{A}$ are negative, i.e., $\tilde{A}$ is negative definite. Furthermore, for any non-negative column vector b with compatible dimensions, if $b \neq 0$, then $A - \text{diag}(b)$ is negative definite.

To establish our first result, it is necessary to give the following assumption for system (6) which is always called as the growth condition [14], [22], [43].

The growth condition (g.c.): for $f \in \mathcal{F}$, there exist constants h_1 and h_2 , with $h_1 \geq 0$ such that

$$\|f(x) - f(y)\| = \sup_{\xi \in \mathbb{F}[f(x)-f(y)]} \|\xi\| \leq h_1 \|x - y\| + h_2. \quad (11)$$

A. FTS With Discontinuous Controllers

Our first goal is to derive the coupled networks (6) synchronize to the isolated neural network (4) with discontinuous controllers. To this end, the controller $u_{\pi i}(t)$ is designed as follows:

$$u_{\pi i}(t) = -\rho_{\pi} \text{sign}(z_i - x), \quad i = 1, 2, \dots, N \quad (12)$$

where $\rho_{\pi} > 0$, for $\pi \in \{1, 2, \dots, M\}$.

Subtracting (4) from (8), we obtain the error dynamical systems

$$\begin{aligned} \dot{e}_i(t) = \sum_{q=1}^M \tilde{h}_q(t) \left\{ -D_q e_i(t) + B_q g_i(t) \right. \\ \left. + c \sum_{j=1}^N a_{qij} \cdot [e_j - e_i + \chi \text{sign}(e_j - e_i)] \right. \\ \left. - \rho_q \text{sign}(e_i) \right\} \end{aligned} \quad (13)$$

where $e_i(t) = z_i(t) - x(t)$ and $g_i(t) = f(z_i(t)) - f(x(t))$, $i = 1, 2, \dots, N$.

1) *Part A1 (Activations are Discontinuous):* First, we consider the FTS problem with the discontinuous activation $f \in \mathcal{F}$. It is worth noting that the existence of solutions for the discontinuous system (13) can be ensured under the framework of Filippov solutions (see [43], [48] for detail).

Theorem 1: Under the g.c. and the controller (12), the coupled networks (6) synchronize to the isolated neural network (4) in a finite time, if for $q = 1, 2, \dots, M$, there exist a series of constants η_q , such that

$$-I_N \otimes D_q - cL_q \otimes I_n + \eta_q I_N \otimes I_n \leq 0 \quad (14)$$

$$\eta_q > n\tilde{b}_q h_1 \quad (15)$$

$$\rho_q > n\tilde{b}_q h_2 \quad (16)$$

where $B_q = (b_{qkl})_{k,l=1}^n$ and $\tilde{b}_q = \max_{1 \leq k, l \leq n} b_{qkl}$.

Proof: For $i = 1, 2, \dots, N$, denote $\delta_i(t)$ as the right hand of the error dynamical systems (13) and $\delta(t) = (\delta_1^T(t), \delta_2^T(t), \dots, \delta_N^T(t))^T$. By Filippov regularization,

we have

$$\begin{aligned} \mathbb{F}[\delta_i(t)] &= \mathbb{F} \left[\sum_{q=1}^M \tilde{h}_q(t) \left\{ -D_q e_i(t) + B_q g_i(t) \right. \right. \\ &\quad \left. \left. + c \sum_{j=1}^N a_{qij} [e_j - e_i + \chi \text{sign}(e_j - e_i)] \right. \right. \\ &\quad \left. \left. - \rho_q \text{sign}(e_i) \right\} \right] \\ &\subseteq \sum_{q=1}^M \tilde{h}_q(t) \left\{ -D_q e_i(t) + B_q \mathbb{F}[g_i](t) \right. \\ &\quad \left. + c \sum_{j=1}^N a_{qij} \mathbb{F}[e_j - e_i + \chi \text{sign}(e_j - e_i)] \right. \\ &\quad \left. - \rho_q \mathbb{F}[\text{sign}(e_i)] \right\} \\ &\subseteq \sum_{q=1}^M \tilde{h}_q(t) \left\{ -D_q e_i(t) + B_q \mathbb{F}[g_i](t) \right. \\ &\quad \left. + c \sum_{j=1}^N a_{qij} [e_j - e_i + \chi \text{SIGN}(e_j - e_i)] \right. \\ &\quad \left. - \rho_q \text{SIGN}(e_i) \right\} \end{aligned} \quad (17)$$

where $\text{SIGN}(e_j - e_i) = (\text{SIGN}(e_{j1} - e_{i1}), \text{SIGN}(e_{j2} - e_{i2}), \dots, \text{SIGN}(e_{jn} - e_{in}))^T$, and

$$\text{SIGN}(s) = \begin{cases} -1, & s < 0 \\ [-1, 1], & s = 0 \\ 1, & s > 0. \end{cases}$$

In the following, we will prove the error system (13) is finite-time stable. To this end, we choose the candidate Lyapunov function

$$V(t) = \sum_{i=1}^N e_i^T e_i = e^T(t) e(t).$$

Since each graph $\mathcal{G}(\mathcal{A}_q)$ is undirected, that is $a_{qij} = a_{qji}$ for all $i, j = 1, 2, \dots, N$ and $q = 1, 2, \dots, M$. Then, by Lemma 3, we get

$$\begin{aligned} \sum_{i,j=1}^N a_{qij} e_i^T \text{SIGN}(e_j - e_i) \\ = \sum_{i,j=1}^N a_{qji} e_j^T \text{SIGN}(e_i - e_j) \\ = -\frac{1}{2} \sum_{i,j=1}^N a_{qij} (e_j - e_i)^T \text{SIGN}(e_j - e_i) \\ = -\frac{1}{2} \sum_{i,j=1}^N \sum_{k=1}^n a_{qij} |e_{jk} - e_{ik}|. \end{aligned} \quad (18)$$

Now, calculating the set-valued Lie derivative of $V(t)$ along the trajectories of the augmented system (13), yields

$$\begin{aligned}
 L_\delta V(t) &= [\partial V(t)]^T \cdot \mathbb{F}[\delta(t)] \\
 &\subseteq \sum_{i=1}^N e_i^T(t) \sum_{q=1}^M \tilde{h}_q(t) \left\{ -D_q e_i(t) + B_q \mathbb{F}[g_i](t) \right. \\
 &\quad \left. + c \sum_{j=1}^N a_{qij} [e_j - e_i] + \chi \text{SIGN}(e_j - e_i) \right. \\
 &\quad \left. - \rho_q e_i^T \text{SIGN}(e_i) \right\} \\
 &= 2 \sum_{q=1}^M \tilde{h}_q(t) \left\{ -\sum_{i=1}^N e_i^T(t) D_q e_i(t) + \sum_{i=1}^N e_i^T(t) B_q \mathbb{F}[g_i](t) \right. \\
 &\quad \left. + c \sum_{i,j=1}^N e_i^T a_{qij} (e_j - e_i) - \rho_q \sum_{i=1}^N e_i^T \text{SIGN}(e_i) \right. \\
 &\quad \left. + c\chi \sum_{i,j=1}^N e_i^T a_{qij} \text{SIGN}(e_j - e_i) \right\} \\
 &= 2 \sum_{q=1}^M \tilde{h}_q(t) \left\{ -\sum_{i=1}^N e_i^T(t) D_q e_i(t) + \sum_{i=1}^N e_i^T(t) B_q \alpha_i(t) \right. \\
 &\quad \left. + c \sum_{i,j=1}^N e_i^T a_{qij} (e_j - e_i) - \rho_q \sum_{i=1}^N \sum_{k=1}^n |e_{ik}| \right. \\
 &\quad \left. - \frac{c\chi}{2} \sum_{i,j=1}^N \sum_{k=1}^n a_{qij} |e_{jk} - e_{ik}| \right\} \quad (19)
 \end{aligned}$$

where $\alpha_i(t) \in \mathbb{F}[g_i](t)$ are measurable selections, $i = 1, 2, \dots, N$.

From the g.c., we have

$$\begin{aligned}
 e_i^T(t) B_q \alpha_i(t) &= \sum_{k,l=1}^n e_{ik}^T(t) b_{qkl} \alpha_{il}(t) \\
 &\leq \tilde{b}_q \sum_{k,l=1}^n |e_{ik}(t)| |\alpha_{il}(t)| \\
 &\leq \tilde{b}_q \sum_{k,l=1}^n |e_{ik}(t)| (h_1 |e_{il}(t)| + h_2) \\
 &\leq \tilde{b}_q h_1 \sum_{k,l=1}^n |e_{ik}(t)| |e_{il}(t)| + \tilde{b}_q h_2 \sum_{k,l=1}^n |e_{ik}(t)|. \quad (20)
 \end{aligned}$$

Note that $L_\delta V(t)$ is in fact singleton valued based on the right-hand of (19). Hence, by (18)–(20), and the

conditions (14)–(16), we get

$$\begin{aligned}
 L_\delta V(t) &\leq 2 \sum_{q=1}^M \tilde{h}_q(t) \\
 &\quad \times \left\{ -\sum_{i=1}^N e_i^T(t) D_q e_i(t) + n\tilde{b}_q h_1 \right. \\
 &\quad \times \sum_{i=1}^N \sum_{k,l=1}^n |e_{ik}(t)| |e_{il}(t)| + n\tilde{b}_q h_2 \sum_{i=1}^N \sum_{k=1}^n |e_{ik}(t)| \\
 &\quad + c \sum_{i,j=1}^N e_i^T a_{qij} (e_j - e_i) - \rho_q \sum_{i=1}^N \sum_{k=1}^n |e_{ik}| \\
 &\quad \left. - \frac{c\chi}{2} \sum_{i,j=1}^N \sum_{k=1}^n a_{qij} |e_{jk} - e_{ik}| \right\} \\
 &= \sum_{q=1}^M \tilde{h}_q(t) \left\{ -2e^T(t) (I_N \otimes D_q) e(t) + 2n\tilde{b}_q h_1 V(t) \right. \\
 &\quad - 2ce^T(t) (L_q \otimes I_n) e(t) \\
 &\quad - c\chi \sum_{i,j=1}^N \sum_{k=1}^n a_{qij} |e_{jk} - e_{ik}| \\
 &\quad \left. - 2(\rho_q - n\tilde{b}_q h_2) \sum_{i=1}^N \sum_{k=1}^n |e_{ik}| \right\} \\
 &= \sum_{q=1}^M \tilde{h}_q(t) \left\{ -2e^T(t) (I_N \otimes D_q) e(t) \right. \\
 &\quad - 2ce^T(t) (L_q \otimes I_n) e(t) \\
 &\quad + 2\eta_q V(t) - (2\eta_q - 2n\tilde{b}_q h_1) V(t) \\
 &\quad - c\chi \sum_{i,j=1}^N \sum_{k=1}^n a_{qij} |e_{jk} - e_{ik}| \\
 &\quad \left. - 2(\rho_q - n\tilde{b}_q h_2) \sum_{i=1}^N \sum_{k=1}^n |e_{ik}(t)| \right\}. \quad (21)
 \end{aligned}$$

By Lemma 2, we obtain

$$\begin{aligned}
 &-c\chi \sum_{i,j=1}^N \sum_{k=1}^n a_{qij} |e_{jk} - e_{ik}| - 2(\rho_q - n\tilde{b}_q h_2) \sum_{i=1}^N \sum_{k=1}^n |e_{ik}(t)| \\
 &= -c\chi \sum_{k=1}^n \left(\sum_{i,j=1}^N a_{qij} |e_{jk} - e_{ik}| \right. \\
 &\quad \left. + \frac{2(\rho_q - n\tilde{b}_q h_2)}{c\chi} \sum_{i=1}^N |e_{ik}(t)| \right) \\
 &\leq -c\chi \left[\sum_{k=1}^n \left(\sum_{i,j=1}^N a_{qij}^2 (e_{jk} - e_{ik})^2 \right. \right. \\
 &\quad \left. \left. + \left(\frac{2\rho_q - 2n\tilde{b}_q h_2}{c\chi} \right)^2 e_{ik}^2(t) \right) \right]^{\frac{1}{2}} \\
 &= -c\chi [e^T(t) ((\tilde{L}_q + \Phi_q) \otimes I_n) e(t)]^{\frac{1}{2}} \quad (22)
 \end{aligned}$$

where $\tilde{L}_q$ is the Laplacian matrix associated with the adjacency matrix $\tilde{A}_q = [a_{qij}^2]_{N \times N}$ and $\Phi_q = ((2\rho_q - 2n\tilde{b}_qh_2)/c\chi)^2 I_N$.

Based on Lemma 4, the matrix $\tilde{L}_q + \Phi_q$ is positive definite and hence the minimum eigenvalue $\lambda_{\min}(\tilde{L}_q + \Phi_q) > 0$.

Then, from the definition of $\tilde{h}_q(t)$ in (7), the condition (14) and Lemma 2, we have

$$\begin{aligned} L_\delta V(t) &\leq \sum_{q=1}^M \tilde{h}_q(t) \left\{ -\left(2\eta_q - 2n\tilde{b}_qh_1\right)V(t) \right. \\ &\quad \left. - c\chi \sqrt{\lambda_{\min}(\tilde{L}_q + \Phi_q)} V^{\frac{1}{2}}(t) \right\} \\ &\leq -\tilde{\eta}V(t) - c\chi\tilde{\lambda}V^{\frac{1}{2}}(t) \end{aligned} \quad (23)$$

where $\tilde{\eta} = \min_{1 \leq q \leq M} (2\eta_q - 2n\tilde{b}_qh_1)$, $\tilde{\lambda} = \min_{1 \leq q \leq M} \sqrt{\lambda_{\min}(\tilde{L}_q + \Phi_q)}$.

By case 1) in Lemma 1, we obtain $V(t) = 0$ for $t \geq \mathbf{T}_1$, where

$$\mathbf{T}_1 = \frac{2}{\tilde{\eta}} \ln \left[1 + \frac{\tilde{\eta}V^{\frac{1}{2}}(0)}{c\chi\tilde{\lambda}} \right] \leq \frac{2}{\tilde{\eta}} \ln \left[1 + \frac{\tilde{\eta}\|e(0)\|}{c\chi\tilde{\lambda}} \right]. \quad (24)$$

Hence, under the control algorithm (12), the nonlinear coupled networks (6) synchronize to the isolated neural network (4) in a finite time $\mathbf{T}_1$. ■

Remark 2: It can be seen from (8) and (12) that the activation functions $f_i(t)$, the controller, and the nonlinear coupling are all discontinuous. Hence, the Filippov regularization should be introduced to make the single-valued function transform into the set-valued one and the corresponding differential equation into the differential inclusion. However, unlike the latter two specific functions (both are signum functions), the activations $f_i(t)$ are unknown in this paper. When calculating the set-valued Lie derivative (19), the measurable selection $\alpha_i(t)$ is further needed to handle such general uncertain discontinuous functions.

Based on case 1) in Lemma 1, we obtained Theorem 1. In the following, we will get another theorem by case 2) in Lemma 1.

Theorem 2: Under the g.c. and the controller (12), the coupled networks (6) synchronize to the isolated neural network (4) in a finite time, if for $q = 1, 2, \dots, M$, the condition (16) and the following inequality hold:

$$-I_N \otimes D_q - cL_q \otimes I_n + n\tilde{b}_qh_1 I_N \otimes I_n \leq 0. \quad (25)$$

Proof: Based on the condition (25) and similar to the proof of Theorem 1, we have (26), as shown at the bottom of this page. By case 2) in Lemma 1, we obtain $V(t) = 0$ for $t \geq \mathbf{T}_2$, where

$$\mathbf{T}_2 = \frac{2V^{\frac{1}{2}}(0)}{c\chi\tilde{\lambda}} \leq \frac{2\|e(0)\|}{c\chi\tilde{\lambda}}. \quad (27)$$

Hence, under the control algorithm (12), the nonlinear coupled networks (6) synchronize to the isolated neural network (4) in a finite time $\mathbf{T}_2$. ■

2) *Part A2 (Activations are Continuous):* It is known that the gain ρ_π is naturally larger than zero in the designed controllers $u_{\pi i}(t)$ (12), then the condition (16) in Theorems 1 and 2 is always holden if h_2 equals to zero in the g.c. However, if $h_2 = 0$ in (9), the g.c. is indeed the following familiar Lipschitz condition:

$$\|f(x) - f(y)\| \leq h_1 \|x - y\|. \quad (28)$$

Hence, in the following, we will continue to discuss the FTS issue for SCNNs under the usual Lipschitz condition (28).

Theorem 3: Under the Lipschitz condition (28) and the controller (12), the coupled networks (6) synchronize to the isolated neural network (4) in a finite time, if for $q = 1, 2, \dots, M$, there exist a series of constants η_q , such that the conditions (14) and (15) hold.

Proof: For the Lipschitz continuous f , the set-valued map $\mathbb{F}(x) = f(x)$ and the measurable selections $\alpha_i(t) = g_i(t)$ in the

$$\begin{aligned} L_\delta V(t) &\leq \sum_{q=1}^M \tilde{h}_q(t) \left\{ -2e^T(t)(I_N \otimes D_q)e(t) + 2n\tilde{b}_qh_1V(t) - 2ce^T(t)(L_q \otimes I_n)e(t) - c\chi \sum_{i,j=1}^N \sum_{k=1}^n a_{qij} |e_{jk} - e_{ik}| \right. \\ &\quad \left. - 2(\rho_q - n\tilde{b}_qh_2) \sum_{i=1}^N \sum_{k=1}^n |e_{ik}| \right\} \\ &= \sum_{q=1}^M \tilde{h}_q(t) \left\{ -2e^T(t)(I_N \otimes D_q)e(t) - 2ce^T(t)(L_q \otimes I_n)e(t) + 2n\tilde{b}_qh_1V(t) - c\chi \sum_{i,j=1}^N \sum_{k=1}^n a_{qij} |e_{jk} - e_{ik}| \right. \\ &\quad \left. - 2(\rho_q - n\tilde{b}_qh_2) \sum_{i=1}^N \sum_{k=1}^n |e_{ik}(t)| \right\} \\ &\leq \sum_{q=1}^M \tilde{h}_q(t) \left\{ -c\chi [e^T(t)((\tilde{L}_q + \Phi_q) \otimes I_n)e(t)]^{\frac{1}{2}} \right\} \leq -c\chi\tilde{\lambda}V^{\frac{1}{2}}(t) \end{aligned} \quad (26)$$

proof of Theorem 1. Hence, based on (19)–(22), we get

$$\begin{aligned}
L_\delta V(t) &\leq 2 \sum_{q=1}^M \tilde{h}_q(t) \left\{ - \sum_{i=1}^N e_i^T(t) D_q e_i(t) + \sum_{i=1}^N e_i^T(t) B_q g_i(t) \right. \\
&\quad + c \sum_{i,j=1}^N e_i^T a_{qij} (e_j - e_i) - \rho_q \sum_{i=1}^N \sum_{k=1}^n |e_{ik}| \\
&\quad \left. - \frac{c\chi}{2} \sum_{i,j=1}^N \sum_{k=1}^n a_{qij} |e_{jk} - e_{ik}| \right\} \\
&\leq \sum_{q=1}^M \tilde{h}_q(t) \left\{ -2e^T(t) (I_N \otimes D_q) e(t) - 2ce^T(t) (L_q \otimes I_n) e(t) \right. \\
&\quad + 2\eta_q V(t) - (2\eta_q - 2n\tilde{b}_q h_1) V(t) \\
&\quad - c\chi \sum_{i,j=1}^N \sum_{k=1}^n a_{qij} |e_{jk} - e_{ik}| \\
&\quad \left. - 2\rho_q \sum_{i=1}^N \sum_{k=1}^n |e_{ik}(t)| \right\}. \quad (29)
\end{aligned}$$

Then, letting $\tilde{\Phi}_q = (2\rho_q/c\chi)^2 I_N$, we obtain

$$\begin{aligned}
L_\delta V(t) &\leq \sum_{q=1}^M \tilde{h}_q(t) \left\{ - (2\eta_q - 2n\tilde{b}_q h_1) V(t) \right. \\
&\quad \left. - c\chi \sqrt{\lambda_{\min}(\tilde{L}_q + \tilde{\Phi}_q)} V^{\frac{1}{2}}(t) \right\} \\
&\leq -\tilde{\eta} V(t) - c\chi \bar{\lambda} V^{\frac{1}{2}}(t) \quad (30)
\end{aligned}$$

where $\bar{\lambda} = \min_{1 \leq q \leq M} \sqrt{\lambda_{\min}(\tilde{L}_q + \tilde{\Phi}_q)}$.

Similarly, by case 1) in Lemma 1, we obtain $V(t) = 0$ for $t \geq \mathbf{T}_3$, where

$$\mathbf{T}_3 = \frac{2}{\tilde{\eta}} \ln \left[1 + \frac{\tilde{\eta} V^{\frac{1}{2}}(0)}{c\chi \bar{\lambda}} \right] \leq \frac{2}{\tilde{\eta}} \ln \left[1 + \frac{\tilde{\eta} \|e(0)\|}{c\chi \bar{\lambda}} \right]. \quad (31)$$

So, for the Lipschitz continuous f , the nonlinear coupled networks (6) synchronize to the isolated neural network (4) in a finite time $\mathbf{T}_3$. ■

Similarly, utilizing case 2) in Lemma 1, we can obtain the following corollary based on Theorems 2 and 3.

Corollary 1: Under the Lipschitz condition (28) and the pinning controller (12), the coupled networks (6) synchronize to the isolated neural network (4) in a finite time, if for $q = 1, 2, \dots, M$, the condition (25) holds. Moreover, the settling time for synchronization is estimated by $\bar{\mathbf{T}}_3 \leq (2\|e(0)\|/c\chi \bar{\lambda})$.

Proof: The proof is similar to that of Theorems 2 and 3 and hence is omitted here. ■

On the other hand, due to network topology is undirected, we now consider the pinning control algorithm with a single controller which is given as follows:

$$\bar{u}_{\pi i}(t) = \begin{cases} -\rho_\pi \text{sign}(z_1 - x), & i = 1 \\ 0, & \text{otherwise.} \end{cases} \quad (32)$$

Theorem 4: Under the Lipschitz condition (28) and the pinning controller (32), the coupled networks (6) synchronize to the isolated neural network (4) in a finite time, if for $q = 1, 2, \dots, M$, there exist a series of constants η_q , such that the conditions (14) and (15) hold.

Proof: Under the pinning controller $\bar{u}_{\pi i}(t)$, the error dynamical systems update to (33), as shown at the bottom of the this page.

By a similar discussion of Theorem 3, we obtain

$$\begin{aligned}
L_\delta V(t) &\leq 2 \sum_{q=1}^M \tilde{h}_q(t) \left\{ - \sum_{i=1}^N e_i^T(t) D_q e_i(t) + \sum_{i=1}^N e_i^T(t) B_q g_i(t) \right. \\
&\quad + c \sum_{i,j=1}^N e_i^T a_{qij} (e_j - e_i) - \rho_q \sum_{k=1}^n |e_{1k}| \\
&\quad \left. - \frac{c\chi}{2} \sum_{i,j=1}^N \sum_{k=1}^n a_{qij} |e_{jk} - e_{ik}| \right\} \\
&\leq \sum_{q=1}^M \tilde{h}_q(t) \left\{ -2e^T(t) (I_N \otimes D_q) e(t) \right. \\
&\quad - 2ce^T(t) (L_q \otimes I_n) e(t) \\
&\quad + 2\eta_q V(t) - (2\eta_q - 2n\tilde{b}_q h_1) V(t) \\
&\quad - c\chi \sum_{i,j=1}^N \sum_{k=1}^n a_{qij} |e_{jk} - e_{ik}| \\
&\quad \left. - 2\rho_q \sum_{k=1}^n |e_{1k}(t)| \right\}. \quad (34)
\end{aligned}$$

By using Lemma 2 again, we obtain

$$\begin{aligned}
&-c\chi \sum_{i,j=1}^N \sum_{k=1}^n a_{qij} |e_{jk} - e_{ik}| - 2\rho_q \sum_{k=1}^n |e_{1k}(t)| \\
&= -c\chi \sum_{k=1}^n \left(\sum_{i,j=1}^N a_{qij} |e_{jk} - e_{ik}| + \frac{2\rho_q}{c\chi} |e_{1k}(t)| \right) \\
&\leq -c\chi \left[\sum_{k=1}^n \left(\sum_{i,j=1}^N a_{qij}^2 (e_{jk} - e_{ik})^2 + \left(\frac{2\rho_q}{c\chi} \right)^2 e_{1k}^2(t) \right) \right]^{\frac{1}{2}} \\
&= -c\chi \left[e^T(t) \left((\tilde{L}_q + \hat{\Phi}_q) \otimes I_n \right) e(t) \right]^{\frac{1}{2}} \quad (35)
\end{aligned}$$

where $\hat{\Phi}_q = (2\rho_q/c\chi)^2 I_1$, and $I_1 = \text{diag}(1, 0, \dots, 0)$.

$$\dot{e}_i(t) = \begin{cases} \sum_{q=1}^M \tilde{h}_q(t) \left\{ -D_q e_i(t) + B_q g_i(t) + c \sum_{j=1}^N a_{qij} \cdot [e_j - e_i + \chi \text{sign}(e_j - e_i)] - \rho_q \text{sign}(e_1) \right\}, & \text{if } i = 1 \\ \sum_{q=1}^M \tilde{h}_q(t) \left\{ -D_q e_i(t) + B_q g_i(t) + c \sum_{j=1}^N a_{qij} \cdot [e_j - e_i + \chi \text{sign}(e_j - e_i)] \right\}, & \text{otherwise} \end{cases} \quad (33)$$

Based on Lemma 4, the matrix $\tilde{L}_q + \hat{\Phi}_q$ is positive definite and $\lambda_{\min}(\tilde{L}_q + \hat{\Phi}_q) > 0$.

Then, combining (34) and (35), we get

$$\begin{aligned} L_\delta V(t) &\leq \sum_{q=1}^M \tilde{h}_q(t) \left\{ -\left(2\eta_q - 2n\tilde{b}_q h_1\right) V(t) \right. \\ &\quad \left. - c\chi \sqrt{\lambda_{\min}(\tilde{L}_q + \hat{\Phi}_q)} V^{\frac{1}{2}}(t) \right\} \\ &\leq -\tilde{\eta} V(t) - c\chi \hat{\lambda} V^{\frac{1}{2}}(t) \end{aligned} \quad (36)$$

where $\hat{\lambda} = \min_{1 \leq q \leq M} \sqrt{\lambda_{\min}(\tilde{L}_q + \hat{\Phi}_q)}$.

By case 1) in Lemma 1, we obtain $V(t) = 0$ for $t \geq \hat{T}_3$, where

$$\hat{T}_3 = \frac{2}{\tilde{\eta}} \ln \left[1 + \frac{\tilde{\eta} V^{\frac{1}{2}}(0)}{c\chi \hat{\lambda}} \right] \leq \frac{2}{\tilde{\eta}} \ln \left[1 + \frac{\tilde{\eta} \|e(0)\|}{c\chi \hat{\lambda}} \right]. \quad (37)$$

So, for the Lipschitz continuous f , the nonlinear coupled networks (6) synchronize to the isolated neural network (4) in a finite time $\hat{T}_3$. ■

Note that the conditions of Theorem 3 and Corollary 1 may be somewhat strong, because the information of every $\tilde{b}_q = \max_{1 \leq k, l \leq n} b_{qkl}$ is needed for each $q = 1, 2, \dots, M$. In practical applications, such theoretical results will lead to more restrictive and conservative, which can be relaxed through the following new corollary.

Corollary 2: Under the Lipschitz condition (28) and the pinning controller (32), the coupled networks (6) synchronize to the isolated neural network (4) in a finite time, if for $q = 1, 2, \dots, M$, there exists a positive constant ε , such that

$$I_N \otimes \left(-D_q + \varepsilon \|B_q\|^2 I_n + \frac{1}{4\varepsilon} h_1^2 I_n \right) - cL_q \otimes I_n \leq 0. \quad (38)$$

Moreover, the settling time for synchronization is estimated by $\hat{T}_3$.

Proof: By a similar discussion as in the proof of Theorem 4, we have

$$\begin{aligned} L_\delta V(t) &\leq 2 \sum_{q=1}^M \tilde{h}_q(t) \left\{ -\sum_{i=1}^N e_i^T(t) D_q e_i(t) + \sum_{i=1}^N e_i^T(t) B_q g(e_i(t)) \right. \\ &\quad + c \sum_{i,j=1}^N e_i^T a_{qij} (e_j - e_i) \\ &\quad - \frac{c\chi}{2} \sum_{i,j=1}^N \sum_{k=1}^n a_{qij} |e_{jk} - e_{ik}| \\ &\quad \left. - \rho_q p_{q1} |e_{1k}| \right\} \end{aligned} \quad (39)$$

where

$$\begin{aligned} e_i^T(t) B_q g(e_i(t)) &\leq \varepsilon e_i^T(t) B_q B_q^T e_i(t) + \frac{1}{4\varepsilon} g^T(e_i(t)) g(e_i(t)) \\ &\leq \varepsilon e_i^T(t) \|B_q\|^2 e_i(t) + \frac{1}{4\varepsilon} h_1^2 e_i^T(t) e_i(t). \end{aligned} \quad (40)$$

Then, by the condition (38), we obtain

$$\begin{aligned} L_\delta V(t) &\leq \sum_{q=1}^M \tilde{h}_q(t) \left\{ -2e^T(t) (I_N \otimes D_q) e(t) \right. \\ &\quad - 2ce^T(t) (L_q \otimes I_n) e(t) \\ &\quad + 2\varepsilon e^T(t) (I_N \otimes \|B_q\|^2 I_n) e(t) \\ &\quad + \frac{2}{4\varepsilon} e^T(t) (I_N \otimes h_1^2 I_n) e(t) \\ &\quad - c\chi \sum_{i,j=1}^N \sum_{k=1}^n a_{qij} |e_{jk} - e_{ik}| \\ &\quad \left. - 2\rho_q \sum_{k=1}^n |e_{1k}(t)| \right\} \\ &\leq -c\chi \hat{\lambda} V^{\frac{1}{2}}(t). \end{aligned} \quad (41)$$

Hence, by case 2) in Lemma 1 and similar to Corollary 1, we obtain $V(t) = 0$ for $t \geq \hat{T}_3$. This completes the proof. ■

B. FTS With Continuous Controllers

In this part, we proceed to consider the FTS issue for linear coupled complex networks with continuous controllers, which is described as follows:

$$\begin{aligned} \dot{z}_i(t) &= -D_\pi z_i(t) + B_\pi f(z_i(t)) + J(t) \\ &\quad + c \sum_{j=1}^N a_{\pi ij} (z_j - z_i) + v_{\pi i}(t) \end{aligned} \quad (42)$$

where $v_{\pi i}(t) = -\rho_\pi \text{sign}(z_i - x) |z_i - x|^\theta$, $1 < \theta < 1$, $i = 1, 2, \dots, N$ are continuous controllers.

Then, the corresponding error system is

$$\begin{aligned} \dot{e}_i(t) &= \sum_{q=1}^M \tilde{h}_q(t) \left\{ -D_q e_i(t) + B_q g(e_i(t)) \right. \\ &\quad \left. + c \sum_{j=1}^N a_{qij} (e_j - e_i) - \rho_q \text{sign}(e_i) |e_i|^\theta \right\}. \end{aligned} \quad (43)$$

1) Part B1 (Activations are Continuous): First, we will consider the FTS problem for SCNNs when activations and controllers are both continuous.

Theorem 5: Under the assumptions of Theorem 3 and with the continuous controllers $v_{\pi i}(t)$, the linear coupled networks (42) synchronize to the isolated neural network (4) in a finite time.

Proof: By a similar discussion as the proof of Theorem 3, we get

$$\begin{aligned} L_\delta V(t) &\leq \sum_{q=1}^M \tilde{h}_q(t) \left\{ -2e^T(t) (I_N \otimes D_q) e(t) \right. \\ &\quad - 2ce^T(t) (L_q \otimes I_n) e(t) \\ &\quad + 2\eta_q V(t) - \left(2\eta_q - 2n\tilde{b}_q h_1\right) V(t) \\ &\quad \left. - 2\rho_q \sum_{i=1}^N \sum_{k=1}^n |e_{ik}(t)|^{\theta+1} \right\}. \end{aligned} \quad (44)$$

Based on Lemma 3, let $\bar{\rho} = \min_{1 \leq q \leq M} \{\rho_q\}$, we obtain

$$\begin{aligned} -2\rho_q \sum_{i=1}^N \sum_{k=1}^n |e_{ik}(t)|^{\theta+1} &\leq -2\rho_q \left(\sum_{k=1}^n \sum_{i=1}^N e_{ik}^2(t) \right)^{\frac{\theta+1}{2}} \\ &= -2\rho_q (e^T(t)e(t))^{\frac{\theta+1}{2}} \\ &\leq -\bar{\rho} V^{\frac{\theta+1}{2}}(t). \end{aligned} \quad (45)$$

Then, we have

$$L_\delta V(t) \leq -\tilde{\eta} V(t) - \bar{\rho} V^{\frac{\theta+1}{2}}(t). \quad (46)$$

By case 1) in Lemma 1 again, we obtain $V(t) = 0$ for $t \geq \mathbf{T}_4$, where

$$\mathbf{T}_4 = \frac{2}{\tilde{\eta}(1-\theta)} \ln \left[1 + \frac{\tilde{\eta} V^{\frac{1-\theta}{2}}(0)}{\bar{\rho}} \right]. \quad (47)$$

Hence, with the continuous controllers $v_{\pi i}(t)$, the linear coupled networks (42) synchronize to the isolated neural network (4) in a finite time $\mathbf{T}_4$. ■

Corollary 3: Under the assumptions of Corollary 1 and with the continuous controllers $v_{\pi i}(t)$, the linear coupled networks (42) synchronize to the isolated neural network (4) in a finite time. Moreover, the settling time for synchronization is estimated by $\tilde{\mathbf{T}}_4 = ((2\tilde{\eta} V^{(1-\theta)/2}(0))/(\bar{\rho}(1-\theta)))$.

Proof: The proof is similar to that of Theorem 5 and Corollary 1 and hence is omitted here. ■

Remark 3: Until now, many research results about the FTS of coupled neural networks were mainly supposed that the activation function f and the designed controller u were both continuous [21]. Very recently, some discontinuous controllers have been introduced to discuss such synchronization problems owing to their immense advantages in designing the finite-time control protocol [22]. In this paper, the issue of FTS is further discussed deeply not only for the discontinuous f via the discontinuous controller u (see Theorems 1–2) but also for continuous f via continuous u (see Theorem 5 and Corollary 3). The obtained results show that the FTS could always be realized never mind the u is continuous or not.

2) *Part B2 (Activations are Discontinuous):* For completeness, in this part we will recall some results that studied before. In [14], we ever claimed that the complete synchronization will not necessary be ensured for the networks with discontinuous activations and continuous controllers. Actually, for $f \in \mathcal{F}$ with the growth condition, the synchronization error may not approach to zero because of $h_2 \neq 0$ which is caused by the discontinuity of f . But the situation will be changed when controllers become discontinuous, and the synchronization error could converge to zero just as discussed in Part A1 of this paper.

Let us return to Theorem 1, we knew that $e_i^T(t)B_q\alpha_i(t) \leq \tilde{b}_q h_1 \sum_{k,l=1}^n |e_{ik}(t)||e_{il}(t)| + \tilde{b}_q h_2 \sum_{k,l=1}^n |e_{ik}(t)|$ from (20). If $h_2 \neq 0$, then the second term in the right hand of (20) is hard to be controlled for the general continuous controllers but can be controlled by discontinuous ones [such as $u_{\pi i}(t)$ in (12)]. Unfortunately, under the continuous controllers, such as $v_{\pi i}(t)$ in (42), it is difficult to compare $|e_{ik}(t)|$ with $|e_{ik}(t)|^{\theta+1}$ based on (44). Hence, in this paper, we insist think that there still

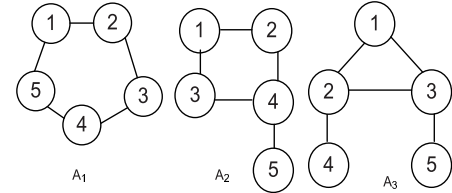


Fig. 1. Three coupled network topologies.

have many difficulties need to overcome when studying the FTS problem if the activations are discontinuous but the controllers are not, which can also be illustrated by one example in the next section of numerical simulations.

Remark 4: In many practical situations, when control accuracy is crucial and stringent convergence time is required, then the synchronization is required to be realized in a finite time. For example, the robotic network tries to agree on the exact value of a quantity sensed by each agent in a finite time, which allows the network to use precise information in the completion of other tasks. But the FTS of SCNNs with discontinuous activations was not completely investigated. In this paper, we deal with the switched systems, discontinuous or/and continuous activations, and finite-time control in a unified framework, which includes and extends the previous researches.

IV. ILLUSTRATIVE EXAMPLES

In this section, one simulation example with four cases are presented to illustrate the utility of theoretical analysis in this paper.

Consider the following switched coupled neural networks model:

$$\begin{aligned} \dot{z}_i(t) &= -D_\pi z_i(t) + B_\pi f(z_i(t)) + J(t) \\ &\quad + c \sum_{j=1}^N a_{\pi ij} [z_j - z_i + \chi \text{sign}(z_j - z_i)] \\ i &= 1, 2, \dots, 5 \end{aligned} \quad (48)$$

where $\pi = 1, 2, 3$; $D_1 = \text{diag}(0.99, 0.97, 0.99)$, $D_2 = \text{diag}(1.01, 1, 0.99)$, and $D_3 = \text{diag}(1, 0.99, 1.01)$; $B_1 = \begin{bmatrix} 3.01 & 0 & 0 \\ 0 & 2 & 1.02 \\ 0 & 1 & 1.99 \end{bmatrix}$, $B_2 = \begin{bmatrix} 2.99 & 0 & 0 \\ 0 & 1.99 & 1.01 \\ 0 & 1 & 2 \end{bmatrix}$, and $B_3 = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0.99 \\ 0 & 1.01 & 2.01 \end{bmatrix}$; and $J(t) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$. Then, $\tilde{b}_1 = 3.01$, $\tilde{b}_2 = 2.99$, and $\tilde{b}_3 = 3$.

A random switching rule is used for the three coupled networks, where the corresponding topologies are illustrated in Fig. 1, where the weight of each edge is 1.

Let $c = 1$. The activation is taken as a discontinuous function $f_d(s) = 0.2s + 0.09\text{sign}(s)$ or a continuous one $f_c(s) = 0.2\tanh(s)$. Then, it is obvious that $h_1 = 0.2$ and $h_2 = 0.18$. The controllers are taken as discontinuous $u_{\pi i}(t) = -\rho_\pi \text{sign}(z_i - x)$ or continuous $v_{\pi i}(t) = -\rho_\pi \text{sign}(z_i - x)|z_i - x|^\theta$.

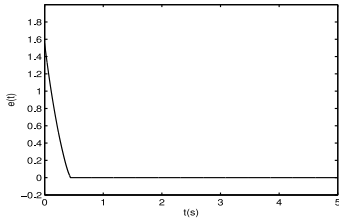


Fig. 2. Evolution of synchronization error versus time for SCNNs with f_d and $u_{\pi i}(t)$.

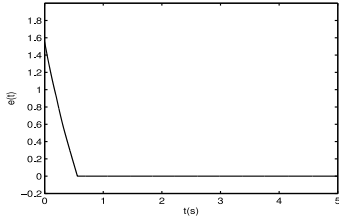


Fig. 3. Evolution of synchronization error versus time for SCNNs with f_c and $u_{\pi i}(t)$.

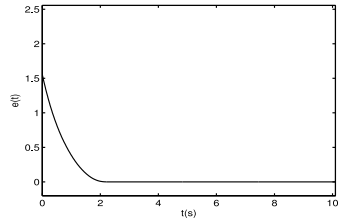


Fig. 4. Evolution of synchronization error versus time for SCNNs with f_c and $v_{\pi i}(t)$.

- 1) *Case A*: For the nonlinear coupled network (48) with $\chi = 0.4$ and the discontinuous activation f_d , by using the discontinuous controllers $u_{\pi i}(t)$ with $\rho_1 = 3.2528$, $\rho_2 = 3.2312$, and $\rho_3 = 3.2420$, the switched network (48) finite-time synchronizes to (4) based on Theorem 2. The evolution of synchronization error $e = \sqrt{\sum_{i=1}^5 e_i^T e_i}$ is shown in Fig. 2. Meanwhile, we can obtain the upper bound of the settling time $T_2 \leq (2\|e(0)\|/c\chi\tilde{\lambda}) = 2.2296$.
- 2) *Case B*: For the nonlinear coupled network (48) with $\chi = 1$ and the continuous activation f_c , by using the discontinuous controllers $u_{\pi i}(t)$ with $\rho_1 = 1.8$, $\rho_2 = 2$, and $\rho_3 = 2.3$, the switched network (48) finite-time synchronizes to (4) based on Corollary 1. The evolution of synchronization error e is shown in Fig. 3, and the upper-bound of the settle time can be obtained as $\bar{T}_3 \leq (2\|e(0)\|/c\chi\tilde{\lambda}) = 2.2762$.
- 3) *Case C*: Now, for the linear coupled network (48) ($\chi = 0$) and the continuous activation f_c , by using the continuous controllers $v_{\pi i}(t)$ with $\rho_1 = \rho_2 = \rho_3 = 1$ and $\theta = 0.5$, the switched network (48) finite-time synchronizes to (4) based on Corollary 3. The evolution of synchronization error e is shown in Fig. 4, and the upper-bound of the settle time can be obtained as $\bar{T}_4 \leq ((2\tilde{\eta}V^{(1-\theta)/2}(0))/(\bar{\rho}(1-\theta))) = 13.6417$.

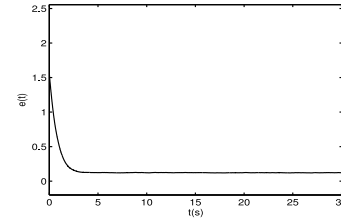


Fig. 5. Evolution of synchronization error versus time for SCNNs with f_d and $v_{\pi i}(t)$.

- 4) *Case D*: In the end, we further consider synchronization issue of SCNNs (48) with the same continuous controllers $v_{\pi i}(t)$ but with discontinuous activation f_d . Based on the discussion in Part B2, the switched network (48) will not synchronize with (4) completely. The evolution of synchronization error e is shown in Fig. 5.

V. CONCLUSION

This paper has introduced a general SCNNs model with discontinuous/continuous activations. Based on the concept of Filippov solution, FTS problems of such networks have been studied by using the nonsmooth analysis approach. Some easy-verified conditions have been established to ensure synchronization for nonlinearly/linearly coupled networks with discontinuous/continuous controllers. Moreover, the corresponding upper bound of the settling time for synchronization can be estimated simultaneously. These results are novel since there are few results on the FTS control of switched networks with both discontinuous activations and controllers. Some numerical examples have been provided to demonstrate the effectiveness of our theoretical results.

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